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# An Invariance Principle for Lattices of Dependent Random Variables

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Dedicated to Professor Leopold Schmetterer

**Summary.** It is shown that for $\varphi$-mixing arrays of Banach space valued random vectors the central limit theorem implies the invariance principle. Applying this result to lattices of random variables a higher dimensional invariance principle under dependence assumptions is obtained.

## 1. Introduction

Under the classical assumptions that the random variables are independent and identically distributed the central limit theorem implies the invariance principle. This fact has been shown by Kuelbs [10] for triangular arrays $\{X_{n,j}\}_{1 \leqq j \leqq n, n \geqq 1}$ of random vectors taking values in a separable Banach space. Garling [6] pushed the result further showing that the entries in each row of the array need not be identically distributed provided second moments are finite and a Lindeberg condition holds. It is the purpose of this paper to drop the other assumption, namely independence and to apply the result obtained to lattices of dependent real-valued random variables.

As we shall see in Sect. 3 basicly the same statement remains true for $\varphi$-mixing arrays. But instead of assuming only convergence of the row sums $X_{n,1} + \ldots + X_{n,n}$ we shall need in general convergence of all subsums $X_{n,1} + \ldots + X_{n,[nu]}$ where $u \in ]0, 1]$. Only if in addition a weak moment condition holds then convergence of the row sums itself will be enough. As a side result coming out of these considerations we shall state slight extensions of two results of Parthasarathy [12] concerning conditional compactness of sequences of probability measures.

In Sect. 4 we consider the stochastic process defined by linear interpolation of partial sums of a lattice of real-valued random variables. We restrict our attention to two-dimensional lattices in order to simplify the notation. Following an idea of Kuelbs [10] we transform the two-dimensional process in an one-

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dimensional one with a Banach space as state space and obtain an invariance principle. This of course means that the process converges in distribution to a Brownian sheet. A similar result has been obtained by Deo [3] for the corresponding process with values in Skorohod space $D_q[0,1]$. His approach is different. He uses results on weak convergence of stochastic processes with multidimensional parameter set. The corresponding theory has been developped in Neuhaus [11], Straf [13], Wichura [14] and Bickel, Wichura [1].

Lattices of dependent random variables are investigated in particular in statistical mechanics. In a recent paper Hegerfeldt, Nappi [8] characterize large classes of lattices which have the dependence structure considered in Sect. 4. Our own motivation for this type of dependence came from random sheets [4]. But the result does not apply there.

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## 2. Notation and Basic Concepts

Throughout this and the next section $E$ will denote a real separable Banach space with norm $\|\cdot\|$ and Borel $\sigma$-algebra $\mathscr{E}$. $E'$ means the topological dual. A measurable mapping $X$ from a probability space $(\Omega, \mathscr{B}, P)$ into $(E, \mathscr{E})$ is called an $E$-valued *random vector*. The distribution of $X$, i.e. the image measure of $P$ under $X$, will be denoted by $P(X)$. If $\{\nu_n\}$ is a sequence of measures on $\mathscr{E}$, $\nu_n \to \nu$ means that $\nu_n$ *converges weakly* to $\nu$. Given a sequence of random vectors $\{X_n\}$ as well as a random vector $X$, $P(X_n) \to P(X)$ will be referred to as $\{X_n\}$ *converges in distribution* to $X$. We say two sequences $\{X_n\}$ and $\{Y_n\}$ of $E$-valued random vectors – defined on the same probability space – are *asymptotically independent* if for each $D \in \mathscr{E} \times \mathscr{E}$

$$\lim_{n\to\infty} |P((X_n, Y_n))(D) - (P(X_n) \otimes P(Y_n))(D)| = 0.$$

Here $\mu \otimes \nu$ denotes the product of two measures $\mu$ and $\nu$. This has to be distinguished from $\mu * \nu$ by which we mean convolution of $\mu$ with $\nu$. Let $\{X_{n,j}\}_{1 \leqq j \leqq n, n \geqq 1}$ be a triangular array of $E$-valued random vectors. If $1 \leqq k \leqq l \leqq n$ write $\mathscr{B}^n_{k,l}$ for the $\sigma$-algebra generated by $\{X_{n,j} | k \leqq j \leqq l\}$. Recall that $\{X_{n,j}\}$ is $\varphi$-*mixing* if $\varphi(k)$ is a function with $\lim_{k\to\infty} \varphi(k) = 0$, such that for events $E_1 \in \mathscr{B}^n_{1,k}$ and $E_2 \in \mathscr{B}^n_{k+l,n}$

$$|P(E_2 | E_1) - P(E_2)| \leqq \varphi(l).$$

The left side is understood to be 0 if $P(E_1) = 0$. Furthermore we can and will assume that $\varphi(k)$ decreases monotonely.

A probability measure $\mu$ on $\mathscr{E}$ is a *mean-zero Gaussian measure* if each $e' \in E'$ has a centered normal distribution. Actually it is enough to verify this for every $e'$ in a linear subspace which separates the points of $E$ (see [6]). Such a measure $\mu$ induces a family $\{\mu_t | t \geqq 0\}$ of mean-zero Gaussian measures which is defined as follows

$$\mu_t(F)=\begin{cases}\delta_0(F) & \text{if } t=0\\ \mu(F/t^{1/2}) & \text{if } t>0.\end{cases}$$

$\{\mu_t \mid t \geqq 0\}$ is a semigroup under convolution since $\mu_s * \mu_t = \mu_{s+t}$. We denote by $C_E$ the space of continuous $E$-valued functions on $[0,1]$ which vanish at 0. Endowed with the norm

$$\|f\|_{C_E} = \sup_{0 \leqq t \leqq 1} \|f(t)\|$$

$C_E$ is a real separable Banach space. Its Borel $\sigma$-algebra $\mathscr{C}$ is generated by the projections $f \to f(t)$. As is well known $\mu$ generates a unique probability measure $W_\mu$ on $\mathscr{C}$ with the following properties: if $0=t_0<t_1<\ldots<t_n$ then the increments $f(t_i)-f(t_{i-1})$ $(i=1,\ldots,n)$ are independent and the distribution of $f(t_i)-f(t_{i-1})$ is $\mu_{t_i-t_{i-1}}$. $W_\mu$ is called the *Wiener measure generated by* $\mu$. The corresponding process $(W_t)_{0 \leqq t \leqq 1}$ defined by $W_t(f)=f(t)$ is *Brownian motion generated by* $\mu$.

For any real number $u$ we denote by $[u]$ the integer part of $u$. $E[X]$ will be written for expectation or mean of a real-valued random variable $X$.

## 3. An Invariance Principle for $E$-valued $\varphi$-mixing Arrays

Given a triangular array of $E$-valued random vectors $\{X_{n,j}\}_{1 \leqq j \leqq n, n \geqq 1}$ we consider the following conditions:

(1) for each $n$ the random vectors $\{X_{n,j}\}_{1 \leqq j \leqq n}$ have the same distribution,

(2) $\{X_{n,j}\}$ is $\varphi$-mixing,

(3) $\{P(u^{-1/2}(X_{n,1}+\ldots+X_{n,[nu]}))\}$ converges weakly to a mean-zero Gaussian measure $\mu$ for each $u \in ]0,1]$,

(4) for each $\varepsilon>0$ and all $h>0$ sufficiently small there exists $\alpha(\varepsilon,h)>\varphi(1)$ such that

$$\varliminf_{n\to\infty} \min_{1 \leqq r \leqq [nh]} P\left[\left\|\sum_{j=1}^{r} X_{n,j}\right\| \leqq \varepsilon\right] \geqq \alpha(\varepsilon,h).$$

We define for each integer $n$ the stochastic process $\{X_n(t)\}_{t\in[0,1]}$ setting

$$X_n(t)=\sum_{j=1}^{[nt]} X_{n,j}+(nt-[nt])\,X_{n,[nt]+1}.$$

Since the paths of this process are in $C_E$, $P(X_n)$ is a measure on $\mathscr{C}$.

(3.1) **Theorem.** *Let $\{X_{n,j}\}_{1 \leqq j \leqq n, n \geqq 1}$ be a triangular array of $E$-valued random vectors satisfying* (1)–(4) *then $\{P(X_n)\}$ converges weakly to Wiener measure $W_\mu$, where $\mu$ is the measure appearing in* (3).

To prove this theorem we will need the next lemma which is given in [9] (Lemma 1.1.6) for real-valued random variables. Our version is proved in the same way.

(3.2) **Lemma.** *Suppose $\{X_{n,j}\}$ is a $\varphi$-mixing array of E-valued random vectors then for any $m \leqq n$ and $t>0$*

$$P\left[\max_{1 \leqq r \leqq m}\left\|\sum_{j=1}^{r} X_{n,j}\right\|>2t\right]<(\alpha-\varphi(1))^{-1} P\left[\left\|\sum_{j=1}^{m} X_{n,j}\right\|>t\right]$$

*whenever* $P\left[\left\|\sum_{j=r+1}^{m} X_{n,j}\right\| \leqq t\right] \geqq \alpha>\varphi(1)$ *for* $r=0,1,\ldots,m-1$.

Note that the random vectors need not be equally distributed but $\varphi(1)<1$ is implicitly required. An immediate consequence of this inequality is that for any fixed $k$ the partial sums $X_{n,1}+\ldots+X_{n,k}$ converge to 0 in probability. More precisely:

(3.3) *Remark.* Let $\{X_{n,j}\}$ be a triangular array satisfying (1)–(4) then for any integer $k$ and any $\varepsilon>0$

$$\lim_{n\to\infty} P\left[\left\|\sum_{j=1}^{k} X_{n,j}\right\|>\varepsilon\right]=0.$$

*Proof.* For some sufficiently small $h>0$ we get

$$\begin{aligned}
&\overline{\lim_{n\to\infty}}\, P\left[\left\|\sum_{j=1}^{k} X_{n,j}\right\|>\varepsilon\right] \\
&\quad \leqq \overline{\lim_{n\to\infty}}\, P\left[\max_{1 \leqq r \leqq [nh]}\left\|\sum_{j=1}^{r} X_{n,j}\right\|>\varepsilon\right] \\
&\quad \leqq \overline{\lim_{n\to\infty}}\, (\alpha(\varepsilon/2,h)-\varphi(1))^{-1} P\left[\left\|\sum_{j=1}^{[nh]} X_{n,j}\right\|>\varepsilon/2\right] \\
&\quad =(\alpha(\varepsilon/2,h)-\varphi(1))^{-1}\, \mu[\|e\|>\varepsilon/2\, h^{1/2}].
\end{aligned}$$

Choosing $h$ small the last term can be made arbitrarily small as is shown more explicitly in the following proof. ⌟

*Proof of* (3.1). The sequence of measures $\{P(X_n)\}$ has a limit point if $\{P(X_n)\}$ is tight. This limit is identified if we show that the finite-dimensional distributions of $P(X_n)$ converge to those of $W_\mu$.

Consider a single time point $t>0$. Since as we just proved $\lim_{n\to\infty} P[\|X_{n,i}\|>\lambda]=0$ for all $i$ and $\lambda>0$ it is enough to consider $\sum_{j=1}^{[nt]} X_{n,j}$ instead of $X_n(t)$ (Theorem 4.1 in [2]). But that $\left\{P\left(\sum_{j=1}^{[nt]} X_{n,j}\right)\right\}$ converges weakly to $W_\mu(W_t)=\mu_t$ follows immediately from (3). Now let $0<r<s<t \leqq 1$ then

$$X_n(t)-X_n(s)\in\mathscr{B}^n_{[ns]+1,[nt]+1} \quad \text{and} \quad X_n(r)\in\mathscr{B}^n_{1,[nr]+1}.$$

From (2) and the argument just used we deduce for sets $F_1, F_2\in\mathscr{E}$

$$\overline{\lim_{n\to\infty}} |P[X_n(r)\in F_1, X_n(t)-X_n(s)\in F_2]-\mu_r(F_1)\mu_{t-s}(F_2)|$$
$$\leqq \overline{\lim_{n\to\infty}} \varphi([n(s-r)]-1)$$
$$+\overline{\lim_{n\to\infty}} |P[X_n(r)\in F_1]P[X_n(t)-X_n(s)\in F_2]-\mu_r(F_1)\mu_{t-s}(F_2)|=0$$

i.e. $\{P((X_n(r), X_n(t)-X_n(s)))\}$ converges weakly to $\mu_r\otimes\mu_{t-s}$. By continuity the same holds for $r=s$ which implies that $\{P((X_n(r), X_n(t)))\}$ converges weakly to $W_\mu((W_r, W_t))$.

More than two time points are treated in the same way. We have only to iterate the inequality occurring in the definition of $\varphi$-mixing.

In order to prove that $\{P(X_n)\}$ is tight it is enough to show that for each $\varepsilon>0$

$$\lim_{h\to 0}\overline{\lim_{n\to\infty}} P[\sup_{|s-t|\leqq h} \|X_n(s)-X_n(t)\|\geqq\varepsilon]=0.$$

Combined with the convergence of the finite-dimensional distributions this allows for $\varepsilon>0$, $\delta>0$ the construction of a set $S_{\varepsilon,\delta}$ which is the union of a finite number of $\delta$-spheres and satisfies $P(X_n)(S_{\varepsilon,\delta})>1-\varepsilon$ for all $n$. By [12] p. 49 this is equivalent to tightness. We proceed as it is done in [12] p. 222 for real-valued random variables and get

$$\overline{\lim_{n\to\infty}} P[\sup_{|s-t|\leqq h} \|X_n(s)-X_n(t)\|>\varepsilon]$$
$$\leqq \sum_{k=0}^{[1/h]} \overline{\lim_{n\to\infty}} P[\sup_{kh\leqq t\leqq (k+1)h} \|X_n(t)-X_n(kh)\|>\varepsilon/4]$$
$$\leqq([1/h]+1)\overline{\lim_{n\to\infty}} P\left[\max_{1\leqq r\leqq [nh]+2}\left\|\sum_{j=1}^{r} X_{n,j}\right\|>\varepsilon/8\right].$$

Writing

$$\left[\max_{1\leqq r\leqq [nh]+2}\left\|\sum_{j=1}^{r} X_{n,j}\right\|>\varepsilon/8\right]$$
$$\subset\left[\max_{1\leqq r\leqq [nh]}\left\|\sum_{j=1}^{r} X_{n,j}\right\|>\varepsilon/8\right]\cup\bigcup_{i=1}^{2}\left[\left\|\sum_{j=1}^{[nh]+i} X_{n,j}\right\|>\varepsilon/8\right]$$

and applying (3.2) to the first term on the right side which is possible by (4) we see that the term above is at most

$$([1/h]+1)((\alpha(\varepsilon/16,h)-\varphi(1))^{-1}+2)$$
$$\cdot\overline{\lim_{n\to\infty}} P\left[\left\|\sum_{j=1}^{[nh]} X_{n,j}\right\|>\varepsilon/16\right]$$
$$=c([1/h]+1)\mu[\|e\|>\varepsilon/16h^{1/2}].$$

$c$ is a constant depending on $\varepsilon$. This expression is with another constant $c'$ bounded by

$$c' \int 1_{[\|e\| > \varepsilon/16h^{1/2}]} \|e\|^2 \, d\mu(e)$$
$$\leqq c' \int 1_{[\|e\| > \varepsilon/16h^{1/2}]} \exp(\lambda \|e\|^2) \, d\mu(e)$$

where $\lambda$ can be chosen arbitrarily small making $h$ small. By a result of Fernique [5] there is a $\lambda > 0$ such that $\int \exp(\lambda \|e\|^2) \, d\mu(e)$ is finite. This implies that the integral above goes to 0 as $h$ goes to 0. ⌟

Assumption (4) implicitly requires $\varphi(1) < 1$. This restriction can be removed at the price of some additional convergence assumption in (3). Suppose $\{X_{n,j}\}$ satisfies (1) and (2) with $\varphi(1) = \ldots = \varphi(k-1) = 1$ and $\varphi(k) < 1$ for some $k > 1$. We define the subarray $\{Y_{n,j}\}_{1 \leqq j \leqq [n/k], n \geqq 1}$ where $Y_{n,j} = X_{n,(j-1)k+1}$ and consider the assumptions

(3a) $\{P(u^{-1/2}(Y_{n,1} + \ldots + Y_{n,[nu/k]}))\}$ converges weakly to a mean-zero Gaussian measure $\nu$ for all $u$ in some interval $]0, u_0]$,

(4′) for each $\varepsilon > 0$ and all $h > 0$ sufficiently small there exists $\beta(\varepsilon, h) > \varphi(k)$ such that

$$\varliminf_{n \to \infty} \min_{1 \leqq r \leqq [nh/k]} P\left[\left\| \sum_{j=1}^{r} Y_{n,j} \right\| \leqq \varepsilon\right] \geqq \beta(\varepsilon, h).$$

Using the following inequality for random vectors $\{V_i\}_{1 \leqq i \leqq n}$ which are sums $V_i = V_i^1 + \ldots + V_i^k$

$$P[\max_{1 \leqq i \leqq n} \|V_i\| > \varepsilon] \leqq \sum_{l=1}^{k} P[\max_{1 \leqq i \leqq n} \|V_i^l\| > \varepsilon/k]$$

the proof of (3.1) can be modified in order to get the following

(3.4) **Theorem.** *Let $\{X_{n,j}\}_{1 \leqq j \leqq n, n \geqq 1}$ be a triangular array of E-valued random vectors satisfying* (1) *and* (2) *with $\varphi(k) < 1$. If assumptions* (3), (3a) *and* (4′) *hold then $\{P(X_n)\}$ converges weakly to Wiener measure $W_\mu$.*

Before turning to an application of (3.1) resp. (3.4) we discuss the question if condition (3) can be weakened. The next proposition shows that this is possible if $\{X_{n,j}\}$ satisfies a weak moment condition. We shall need the following lemma.

(3.5) **Lemma.** *Let $X$, $Y$ be E-valued random vectors. If*

$$|P(X^{-1}A \cap Y^{-1}B) - P(X^{-1}A)P(Y^{-1}B)| \leqq \varepsilon P(X^{-1}A)$$

*for all $A, B \in \mathscr{E}$ then*

$$|P((X,Y))(D) - (P(X) \otimes P(Y))(D)| \leqq \varepsilon$$

*holds for all $D \in \mathscr{E} \times \mathscr{E}$.*

*Proof.* We show first that the conclusion is true for sets $D$ in the algebra generated by rectangles $A \times B$ where $A, B \in \mathscr{E}$. Since a (set-theoretic) difference of two rectangles can be written as a finite union of disjoint rectangles it suffices to consider this family. Now a disjoint union $\bigcup_{i=1}^{n} (\tilde{A}_i \times \tilde{B}_i)$ can always be written as $\bigcup_{i=1}^{m} (A_i \times B_i)$ such that $A_i \cap A_j = \emptyset$ for $i \neq j$. Therefore

$$\left|P((X,Y))\left(\bigcup_{i=1}^{m}(A_i\times B_i)\right)-(P(X)\otimes P(Y))\left(\bigcup_{i=1}^{m}(A_i\times B_i)\right)\right|$$
$$\leqq\sum_{i=1}^{m}|P(X^{-1}A_i\cap Y^{-1}B_i)-P(X^{-1}A_i)P(Y^{-1}B_i)|$$
$$\leqq\varepsilon\sum_{i=1}^{m}P(X^{-1}A_i)\leqq\varepsilon.$$

Obviously the sets $D$ satisfying the conclusion form a monotone class. By [7] p. 26 the conclusion is true for all $D$ in the $\sigma$-algebra generated by the rectangles which is $\mathscr{E}\times\mathscr{E}$. ⌟

Note that the conclusion of (3.5) implies for all $A\in\mathscr{E}$

$$|(P(X+Y))(A)-(P(X)*P(Y))(A)|\leqq\varepsilon.$$

(3.6) **Proposition.** *Let $\{X_{n,j}\}_{1\leqq j\leqq n, n\geqq 1}$ be a triangular array of E-valued random vectors satisfying* (1) *and* (2). *If $\lim_{n\to\infty}E[\|X_{n,1}\|]=0$ then the weak convergence of $\{P(X_{n,1}+\ldots+X_{n,n})\}$ to a mean-zero Gaussian measure $\mu$ implies the weak convergence of $\{P(X_{n,1}+\ldots+X_{n,[nu]})\}$ to $\mu_u$ for every $u\in]0,1]$.*

*Proof.* By choosing a subsequence if necessary we may suppose that $E[\|X_{n,1}\|]\leqq n^{-1}$ for each $n$. First we show that

$$\{P(\sum_{k/2^r<j/n\leqq(k+1)/2^r}X_{n,j})\}\text{ converges to }\mu_{1/2^r}$$

for any $k=0,\ldots,2^r-1$ and any $r=0,1,2,\ldots$. For $r=0$ and $k=0$ this statement is true by assumption. Suppose it is true for $r-1$ and fix $k$ which will be assumed to be 0. Then

$$\{P(\sum_{1\leqq j\leqq n/2^{r-1}}X_{n,j})\}\text{ converges to }\mu_{1/2^{r-1}}.$$

We define

$$R(n)=\{j\,|\,1\leqq j\leqq(n-\sqrt{n})\,2^{-r}\}$$

and

$$Q(n)=\{j\,|\,(n+\sqrt{n})\,2^{-r}<j\leqq 2n\,2^{-r}\}$$

and put $\tilde{X}_n=\sum_{j\in R(n)}X_{n,j}$, $\tilde{Y}_n=\sum_{j\in Q(n)}X_{n,j}$. Neglecting eventually the first $j$ appearing in $Q(n)$ we can achieve that $R(n)$ and $Q(n)$ have the same cardinality. Consequently by (1) $P(\tilde{X}_n)$ equals $P(\tilde{Y}_n)$ for each $n$. Now for $\varepsilon>0$

$$P[\|(\tilde{X}_n+\tilde{Y}_n)-\sum_{1\leqq j\leqq n/2^{r-1}}X_{n,j}\|>\varepsilon]$$
$$\leqq\varepsilon^{-1}E[\|\sum_{1\leqq j\leqq 2\sqrt{n}/2^r}X_{n,j}\|]\leqq 1/\varepsilon\,2^{r-1}\,n^{1/2}$$

which implies that $\{P(\tilde{X}_n+\tilde{Y}_n)\}$ converges to $\mu_{1/2^{r-1}}$ too. On the other side by condition (2)

$$|P(\tilde{X}_n^{-1}A\cap\tilde{Y}_n^{-1}B)-P(\tilde{X}_n^{-1}A)P(\tilde{Y}_n^{-1}B)|\leqq\varphi([\sqrt{n}2^{-r+1}])P(\tilde{X}_n^{-1}A)$$

holds for all $A, B\in\mathscr{E}$. Lemma (3.5) implies that $\{\tilde{X}_n\}$ and $\{\tilde{Y}_n\}$ are asymptotically independent. For any $A\in\mathscr{E}$

$$\lim_{n\to\infty}|(P(\tilde{X}_n)*P(\tilde{Y}_n))(A)-P(\tilde{X}_n+\tilde{Y}_n)(A)|=0$$

which means that $\{P(\tilde{X}_n)*P(\tilde{Y}_n)\}$ converges to $\mu_{1/2^{r-1}}$. In the same way as in Kuelbs [10] p. 164 below we see now that $\{P(\tilde{X}_n)\}$ (and $\{P(\tilde{Y}_n)\}$ as well) converges to $\mu_{1/2^r}$. Observing

$$P[\|\sum_{1\leqq j\leqq n/2^r}X_{n,j}-\tilde{X}_n\|>\varepsilon]\leqq 1/\varepsilon 2^r n^{1/2}$$

we see that $\{P(\sum_{1\leqq j\leqq n/2^r}X_{n,j})\}$ converges to $\mu_{1/2^r}$ which finishes the induction step for the statement at the beginning of the proof. Now we proceed to show that

$$\{P(\sum_{j/n\leqq k/2^r}X_{n,j})\}\text{ converges to }\mu_{k/2^r}$$

for any $r=0,1,2,\ldots$ and $k=1,\ldots,2^r$. Let $r>1$ be given then this is contained in the statement just proved for $k=1$ or $k=2$. Suppose it is true for $k-1$. Using the same technique as above - namely introducing asymptotically independent subsums $\tilde{X}_n$ and $\tilde{Y}_n$ - we show that

$$\begin{aligned}&\{P(\sum_{j/n\leqq k/2^r}X_{n,j})\}\\&\quad=\{P(\sum_{j/n\leqq(k-1)/2^r}X_{n,j}+\sum_{(k-1)/2^r<j/n\leqq k/2^r}X_{n,j})\}\end{aligned}$$

converges to $\mu_{(k-1)/2^r}*\mu_{1/2^r}=\mu_{k/2^r}$. The proposition now follows by continuity since we have shown the convergence for all $u$ out of a dense subset of $[0,1]$. ⌟

As an aside let us finally mention two results which in our context extend slightly Theorems 2.1 and 2.2 in Parthasarathy [12] p. 58. Both theorems are rather basic for [12].

(3.7) **Proposition.** *Let $\{X_n\}$ and $\{Y_n\}$ be asymptotically independent sequences of E-valued random vectors. If the sequence $\{P(X_n+Y_n)\}$ is conditionally compact then the sequences of distributions $\{P(X_n)\}$ and $\{P(Y_n)\}$ are shift-compact.*

*Proof.* Given $\varepsilon>0$ there is by assumption a compact set $K_\varepsilon$ such that $P(X_n+Y_n)(K_\varepsilon)>1-\varepsilon/2$ for all $n$. Choose $n_0=n_0(\varepsilon)$ such that for all $n\geqq n_0$ $|(P(X_n+Y_n))(K_\varepsilon)-(P(X_n)*P(Y_n))(K_\varepsilon)|<\varepsilon/2$ then $(P(X_n)*P(Y_n))(K_\varepsilon)>1-\varepsilon$ for all $n\geqq n_0$. Since $E$ is separable and complete each $P(X_n)*P(Y_n)$ $(1\leqq n\leqq n_0)$ is a tight measure. Consequently $\{P(X_n)*P(Y_n)\}$ is conditionally compact and we can apply Theorem 2.2 [12] p. 59. ⌟

Introducing the same approximation argument directly in the proof of Theorem 2.1 [12] p. 58 we see

(3.8) **Proposition.** *Let $\{X_n\}$ and $\{Y_n\}$ be asymptotically independent sequences of E-valued random vectors. If the sequences of distributions $\{P(X_n+Y_n)\}$ and $\{P(X_n)\}$ are conditionally compact then so is $\{P(Y_n)\}$.*

## 4. Application to a Lattice of Random Variables

Denote by $C_0^2$ the space of real-valued continuous functions on $[0,1]\times[0,1]$ which vanish at all points $(s,0)$ and $(0,t)$ for $s,t\in[0,1]$. With the supremum norm $C_0^2$ is a separable Banach space.

Let $\xi=\{\xi_{ij}\}_{1\leqq i,j<\infty}$ be a lattice of real-valued random variables defined on a probability space $(\Omega,\mathscr{B},P)$. We define a sequence $\{Y_n\}_{n\geqq 1}$ of $C_0^2$-valued random vectors as follows

$$\begin{aligned}Y_n(t,u)=n^{-1}\Bigg[&\sum_{l=1}^{[nt]}\sum_{j=1}^{[nu]}\xi_{lj}+\sum_{l=1}^{[nt]}(nu-[nu])\,\xi_{l,[nu]+1}\\&+\sum_{j=1}^{[nu]}(nt-[nt])\,\xi_{[nt]+1,j}\\&+(nu-[nu])(nt-[nt])\,\xi_{[nt]+1,[nu]+1}\Bigg].\end{aligned}$$

If a sum $\sum_{l=1}^{0}$ occurs it is understood to be 0.

(4.1) **Lemma.** *Let $\xi=\{\xi_{ij}\}$ be a lattice of identically distributed random variables of zero expectation. If $\sum_{l,k=1}^{\infty}|E[\xi_{11}\,\xi_{lk}]|<\infty$ and $\sum_{l,k=2}^{\infty}|E[\xi_{l1}\,\xi_{1k}]|<\infty$ then*

$$\lim_{n\to\infty}E[Y_n^2(1,1)]=2\left(\sum_{l,k=1}^{\infty}E[\xi_{11}\,\xi_{lk}]+\sum_{l,k=2}^{\infty}E[\xi_{l1}\,\xi_{1k}]\right)\cdot E[\xi_{11}^2]$$

*and the following uniform bound holds for all $t,u\in[0,1]$*

$$E[Y_n^2(t,u)]\leqq 2\left(\sum_{l,k=1}^{\infty}|E[\xi_{11}\,\xi_{lk}]|+\sum_{l,k=2}^{\infty}|E[\xi_{l1}\,\xi_{1k}]|\right).$$

The proof is done by elementary computations. We shall write $\sigma^2=\sigma_\xi^2=\lim_{n\to\infty}E[Y_n^2(1,1)]$. Under the assumptions made above $\sigma^2$ is a finite non-negative value. $\sigma^2=0$ is of no interest since in this case $Y_n(t,u)$ converges to 0 in probability for every $t,u\in[0,1]$. If $\sigma^2>0$ define

$$X_n(t,u)=\sigma^{-1}\,Y_n(t,u).$$

The next proposition gives a simple sufficient condition that guarantees finiteness of the series in (4.1). For $1\leqq k\leqq l<\infty$ we denote by ${}_1\mathscr{B}_{k,l}$ the $\sigma$-algebra

generated by $\{\xi_{ij} | k \leqq i \leqq l, j \geqq 1\}$ and by ${}_2\mathscr{B}_{m,n}$ the $\sigma$-algebra generated by $\{\xi_{ij} | m \leqq j \leqq n, i \geqq 1\}$ where $1 \leqq m \leqq n < \infty$. It is clear what ${}_1\mathscr{B}_{k,\infty}$, ${}_2\mathscr{B}_{m,\infty}$ means. The lattice $\xi$ is called to be *$\varphi_1$-mixing in the 1-direction* if for events $E_1 \in {}_1\mathscr{B}_{1,k}$, $E_2 \in {}_1\mathscr{B}_{k+n,\infty}$

$$|P(E_2|E_1) - P(E_2)| \leqq \varphi_1(n)$$

and the function $\varphi_1$ satisfies $\lim_{n\to\infty} \varphi_1(n) = 0$. Again the left side is understood to be 0 if $P(E_1) = 0$. $\xi$ is *$\varphi_2$-mixing in the 2-direction* if the same holds with $\varphi_1$ replaced by $\varphi_2$ and ${}_1\mathscr{B}_{1,k}$, ${}_1\mathscr{B}_{k+n,\infty}$ by ${}_2\mathscr{B}_{1,k}$ and ${}_2\mathscr{B}_{k+n,\infty}$. $\xi$ is *$(\varphi_1, \varphi_2)$-mixing* if it is $\varphi_1$-mixing in the 1-direction and $\varphi_2$-mixing in the 2-direction.

(4.2) **Proposition.** *Let $\xi = \{\xi_{ij}\}$ be a lattice of identically distributed random variables of zero expectation and finite variance that is $(\varphi_1, \varphi_2)$-mixing. Then $\sum_{k=1}^{\infty} k\varphi_i(k)^{1/2} < \infty$ for $i = 1, 2$ implies $\sum_{l,k=1}^{\infty} |E[\xi_{11}\xi_{lk}]| < \infty$ as well as $\sum_{l,k=2}^{\infty} |E[\xi_{l1}\xi_{1k}]| < \infty$.*

The proof is immediate if we apply inequality (20.23) of [2]. $C_0^2$ supports two-dimensional Wiener measure $W^{(2)}$ which can be characterized as follows. Let $Z = (Z_u)_{u\in[0,1]^2}$ be a *Brownian sheet*, i.e. a mean-zero Gaussian process with continuous paths and covariance function $E[Z_u Z_v] = \min(u_1, v_1)\min(u_2, v_2)$ where $u = (u_1, u_2)$, $v = (v_1, v_2)$, then $P(Z) = W^{(2)}$.

(4.3) **Theorem.** *Let $\xi = \{\xi_{ij}\}_{1\leqq i,j<\infty}$ be a lattice of identically distributed random variables having mean zero and finite variance. Suppose $\xi$ is $(\varphi_1, \varphi_2)$-mixing such that $\sum_{k=1}^{\infty} k\varphi_i(k)^{1/2} < \infty$ for $i = 1, 2$ and $\sigma_\xi^2 > 0$. Then $\{P(X_n)\}$ converges weakly to $W^{(2)}$.*

To prove this result we apply Theorem (3.1). Thus we have to introduce a suitable array of random vectors. For $j = 1, \ldots, n$ and $t \in [0, 1]$ put

$$X_{n,j}(t) = (\sigma n)^{-1} \left( \sum_{l=1}^{[nt]} \xi_{lj} + (nt - [nt])\, \xi_{[nt]+1,j} \right).$$

$X_{n,j}$ is a $C[0,1]$-valued random vector and if $u \in [0, 1]$

$$X_n(t, u) = \sum_{j=1}^{[nu]} X_{n,j}(t) + (nu - [nu])\, X_{n,[nu]+1}(t).$$

In particular $\{X_n(\cdot, u)\}_{u\in[0,1]}$ is the stochastic process which was denoted $\{X_n(u)\}_{u\in[0,1]}$ in Sect. 3. We have to verify that conditions (1)–(4) hold. The crucial part is (3) which we prepare now by an auxiliary result.

(4.4) **Lemma.** *Let $\xi = \{\xi_{ij}\}$ be a lattice of identically distributed random variables, $|\xi_{11}| \leqq C$, $E[\xi_{11}] = 0$. Suppose $\xi$ is $(\varphi_1, \varphi_2)$-mixing such that $\sum_{k=1}^{\infty} k\varphi_i(k)^{1/2} < \infty$ for $i = 1, 2$. Then for all $t, u \in [0, 1]$*

$$E[Y_n^4(t,u)] \leqq K(\varphi_1, \varphi_2)\, C^4$$

*where $K(\varphi_1, \varphi_2)$ is a constant depending on $\varphi_1$ and $\varphi_2$ only.*

*Proof.* It will be enough to consider the case $t=u=1$. Since if $t$ or $u$ is less than 1 $Y_n(t,u)$ is a sum of the form $t_1\,\xi_1+\ldots+t_m\,\xi_m$ where $0 \leqq t_i \leqq 1$ for all $i$ and $t_i<1$ for some $i$. But

$$E[(t_1\,\xi_1+\ldots+t_m\,\xi_m)^4] \leqq \sum_{\substack{i_1,\ldots,i_m \geqq 0 \\ i_1+\ldots+i_m=4}} \binom{4}{i_1,\ldots,i_m} |E[\xi_1^{i_1}\ldots\xi_m^{i_m}]|.$$

The last sum is actually the term for which we give an upper bound. $\xi_{ij}$ are identically distributed random variables, therefore

$$E[Y_n^4(1,1)] \leqq n^{-4}\,4!\,n^2 \Bigg[ \sum_{\tau\in\mathscr{S}_4} \sum_{\substack{i,j,k\geqq 0 \\ i+j+k\leqq n-1}} \sum_{\substack{l,p,q\geqq 0 \\ l+p+q\leqq n-1}} |E[\xi_{1,m(\tau(1))}\,\xi_{1+i,m(\tau(2))}\,\xi_{1+i+j,m(\tau(3))}\,\xi_{1+i+j+k,m(\tau(4))}]| \Bigg]$$

where $m(1)=1$, $m(2)=1+l$, $m(3)=1+l+p$, $m(4)=1+l+p+q$ and $\mathscr{S}_4$ stands for the permutations of $\{1,\ldots,4\}$. Now the subsum corresponding to $\tau$ being the identity can be written as

$$\begin{aligned} n^{-2}\,4!\,[ & \sum_{j,k\leqq i}\sum_{p,q\leqq l} + \sum_{i,j\leqq k}\sum_{p,q\leqq l} + \sum_{i,k\leqq j}\sum_{p,q\leqq l} \\ & + \sum_{j,k\leqq i}\sum_{p,l\leqq q} + \sum_{i,j\leqq k}\sum_{p,l\leqq q} + \sum_{i,k\leqq j}\sum_{p,l\leqq q} \\ & + \sum_{j,k\leqq i}\sum_{l,q\leqq p} + \sum_{i,j\leqq k}\sum_{l,q\leqq p} + \sum_{i,k\leqq j}\sum_{l,q\leqq p} ] \end{aligned}$$

where the summands which we omitted are those from above and the general restrictions $i,j,k\geqq 0$, $i+j+k\leqq n-1$ as well as $l,p,q\geqq 0$, $l+p+q\leqq n-1$ are kept.

Write $\Sigma_1,\ldots,\Sigma_9$ for these sums. We continue to split

$$\Sigma_1 = \sum_{j,k\leqq i}\sum_{\substack{p,q\leqq l \\ l\leqq i}} + \sum_{j,k\leqq i}\sum_{\substack{p,q\leqq l \\ l>i}} = \Sigma_{11}+\Sigma_{12}.$$

By inequality (20.23) in [2] we have for $i\geqq 1$

$$|E[\xi_{11}(\xi_{1+i,l+1}\,\xi_{1+i+j,1+l+p}\,\xi_{1+i+j+k,1+l+p+q})]| \leqq 2\,C^4\,\varphi_1(i).$$

But the inequality holds for $i=0$ too if we define $\varphi_1(0)=1$. We shall use in the following lines the inequality

$$\sum_{i=0}^{n-1}(i+1)^3\,\varphi_1(i) \leqq 2\sum_{i=0}^{n-1}(i+1)\sum_{u=i}^{n-1}(u+1)\,\varphi_1(u)$$

which can be verified by induction on $n$. We conclude

$$\begin{aligned} n^{-2}\Sigma_{11} &\leqq n^{-2}\sum_{i=0}^{n-1}\sum_{j,k=0}^{i}\sum_{l=0}^{i}\sum_{p,q=0}^{l} 2C^4\varphi_1(i) \\ &\leqq 2C^4 n^{-2}\sum_{i=0}^{n-1}\sum_{j,k=0}^{i}(i+1)^3\varphi_1(i) \\ &\leqq 4C^4\sum_{i=0}^{n-1}(i+1)\varphi_1(i)^{1/2}\sum_{u=i}^{n-1}(u+1)\varphi_1(u)^{1/2} \\ &\leqq 4C^4\left(\sum_{i=0}^{\infty}(i+1)\varphi_1(i)^{1/2}\right)^2. \end{aligned}$$

$\Sigma_{12}$ is treated in the same way, the only difference being that $\varphi_2$ enters instead of $\varphi_1$. $\Sigma_2$, $\Sigma_4$ and $\Sigma_5$ are of the same structure as $\Sigma_1$ thus yield the same bounds.

$$\Sigma_3 = \sum_{i,k\leqq j}\sum_{\substack{p,q\leqq l\\ l\leqq j}} + \sum_{i,k\leqq j}\sum_{\substack{p,q\leqq l\\ l>j}} = \Sigma_{31}+\Sigma_{32}.$$

Again by (20.23) in [2]

$$\begin{aligned} &|E[(\xi_{11}\xi_{1+i,1+l})(\xi_{1+i+j,1+l+p}\xi_{1+i+j+k,1+l+p+q})]| \\ &\quad\leqq |E[\xi_{11}\xi_{1+i,1+l}]|\,|E[\xi_{11}\xi_{1+k,1+q}]| + 2C^4\varphi_1(j) \end{aligned}$$

and

$$\begin{aligned} &|E[\xi_{11}\xi_{1+i,1+l}]| \leqq 2C^2\min\{\varphi_1(i),\varphi_2(l)\}, \\ &|E[\xi_{11}\xi_{1+k,1+q}]| \leqq 2C^2\min\{\varphi_1(k),\varphi_2(q)\}. \end{aligned}$$

Thus

$$\begin{aligned} \Sigma_{31} &\leqq \sum_{i,k\leqq j}\sum_{\substack{p,q\leqq l\\ l\leqq j}} 4C^4\min\{\varphi_1(i),\varphi_2(l)\}\min\{\varphi_1(k),\varphi_2(q)\} \\ &\quad + \sum_{i,k\leqq j}\sum_{\substack{p,q\leqq l\\ l\leqq j}} 2C^4\varphi_1(j). \end{aligned}$$

The second term on the right side has exactly the form of $\Sigma_1$ and thus leads to the same bounds. The first term can be split into $\Sigma_{311}+\Sigma_{312}+\Sigma_{313}+\Sigma_{314}$ where in $\Sigma_{311}$ we sum over all $p\leqq i$, $q\leqq k$, in $\Sigma_{312}$ over all $p\leqq i$, $q>k$, in $\Sigma_{313}$ over all $p>i$, $q\leqq k$ and in $\Sigma_{314}$ over all $p>i$, $q>k$.

$$\begin{aligned} n^{-2}\Sigma_{311} &\leqq 4C^4 n^{-2}\sum_{j=0}^{n-1}\sum_{i,k=0}^{j}\sum_{l=0}^{j}(i+1)\varphi_1(i)(k+1)\varphi_1(k) \\ &\leqq 4C^4\left(\sum_{i=0}^{\infty}(i+1)\varphi_1(i)^{1/2}\right)^2. \end{aligned}$$

Taking the appropriate values out of $\{\varphi_1(i),\varphi_2(l)\}$, $\{\varphi_1(k),\varphi_2(q)\}$ the other three subsums of $\Sigma_{31}$ are treated similarly. $\Sigma_{32}$ can be estimated in the same way as $\Sigma_{12}$. Since $\Sigma_6$, $\Sigma_7$ and $\Sigma_8$ have the structure of $\Sigma_3$ it remains to consider $\Sigma_9$.

Splitting $\Sigma_9$ in $\Sigma_{91}$ nd $\Sigma_{92}$ we can proceed in both cases as we did for $\Sigma_{31}$. This finishes the proof for the subsum corresponding to $\tau$ being the identity. But any other $\tau \in \mathscr{S}_4$ can be treated along the same lines. ⌟

(4.5) **Proposition.** *Let $\xi = \{\xi_{ij}\}_{1 \leqq i,j < \infty}$ be a lattice of identically distributed random variables, $E[\xi_{11}] = 0$ and $E[\xi_{11}^2] < \infty$. Suppose $\xi$ is $(\varphi_1, \varphi_2)$-mixing such that $\sum_{k=1}^{\infty} k \varphi_i(k)^{1/2} < \infty$ for $i = 1, 2$ and $\sigma_\xi > 0$. Then for every fixed $u \in ]0, 1]$ the sequence $\{P(u^{-1/2} X_n(\cdot, [nu]/n))\}$ converges weakly to Wiener measure $W = W^{(1)}$ on $C[0, 1]$.*

*Proof.* We shall verify the sufficient conditions given by Theorem 19.2. [2] and follow closely the proof of Theorem 20.1 [2]. Asymptotic independence of increments can be shown using that $\xi$ is $\varphi_1$-mixing in the 1-direction. The moment conditions are obvious. To see that $\{u^{-1} X_n^2(t, [nu]/n)\}_{n \geqq 1}$ is uniformly integrable for each $t$ we show more namely

$$\lim_{\alpha \to \infty} \sup_{n \geqq 1} \sup_{t \in [0,1]} \int 1_{[|u^{-1} X_n^2(t, [nu]/n)| > \alpha]} u^{-1} X_n^2(t, [nu]/n)\, dP = 0.$$

If $|\xi_{11}| \leqq C$ then we can apply Lemma (4.4) since the integral is at most

$$\alpha^{-1} u^{-2} \int X_n^4(t, [nu]/n)\, dP$$

and get the upper bound

$$\alpha^{-1} u^{-2} (C/\sigma)^4 K(\varphi_1, \varphi_2).$$

From this the result follows in the case considered. If $|\xi_{11}|$ is not bounded we proceed as on p. 176 [2] making some obvious modifications.

It remains to verify the tightness condition. Adapting the proof of Theorem 8.4 [2] we see that it is enough to show the following: for all $\varepsilon > 0$ there exists a $\lambda > 1$ and an integer $n_0$ such that if $n \geqq n_0$ then

$$P\left[\max_{i \leqq n} \left| \sum_{j=1}^{[nu]} \sum_{l=1}^{i} \xi_{lj} \right| \geqq \lambda \sigma n\right] \leqq \varepsilon/\lambda.$$

To this end we first show that there exists a strictly increasing sequence $m_i$ such that $n \geqq m_i$ implies

$$P\left[\left| \sum_{j=1}^{[nu]} \xi_{1j} \right| \geqq \sigma n i^{-2}\right] \leqq 1/n i^2.$$

Write $S_l = \sum_{j=1}^{l} \xi_{1j}$. Assume there exists an increasing sequence $\{n_k\}_{k \geqq 1}$ such that for all $k$

$$P[|S_{[n_k u]}| \geqq \sigma n_k i^{-2}] > 1/n_k i^2.$$

Then

$$\begin{aligned} \sigma^2 i^{-6} &< \sigma^2 n_k i^{-4} P[|S_{[n_k u]}| \geqq \sigma n_k i^{-2}] \\ &\leqq \int 1_{[n_k^{-1/2} |S_{[n_k u]}| \geqq \sigma n_k^{1/2} i^{-2}]} n_k^{-1} S_{[n_k u]}^2\, dP. \end{aligned}$$

We produce a contradiction if we show that the last integral converges to 0. Since $\{\xi_{1j}\}_{j\geqq 1}$ is $\varphi_2$-mixing with $\sum_{k=1}^{\infty} \varphi_2(k)^{1/2} < \infty$ Lemma 3, p. 172 [2] and the lines following it imply that $E[n^{-1} S_{[nu]}^2]$ converges. Furthermore by Theorem 20.1 [2] $\{n^{-1/2} S_{[nu]}\}$ converges in distribution to a normally distributed random variable - which may be degenerate. Both facts together imply that the integral above goes to 0 as $k$ goes to infinity. Now write $p_n = i$ for $m_i \leqq n \leqq m_{i+1}$, then $p_n$ goes to infinity and

$$\lim_{n\to\infty} nP[U_{p_n}^* \geqq \lambda \sigma n] = 0$$

for all $\lambda > 0$, where $U_p^* = \sum_{k=1}^{p} \left| \sum_{j=1}^{[nu]} \xi_{kj} \right|$.

With these preparations the rest is accomplished using the arguments on p. 175 [2]. ⌟

*Proof of Theorem* (4.3). Assumptions (1) and (2) in Theorem (3.1) are obvious. (3) is what we proved as (4.5). The mean-zero Gaussian measure is Wiener measure $W^{(1)}$. Now suppose $\varphi_2(1) < 1$ at first. Given $\varepsilon > 0$ and $h > 0$ sufficiently small we have for $1 \leqq r \leqq [nh]$

$$P\left[\left\| \sum_{j=1}^{r} X_{n,j} \right\| > \varepsilon\right] = P\left[\max_{1\leqq i\leqq n} \left| (\sigma n)^{-1} \sum_{j=1}^{r} \sum_{l=1}^{i} \xi_{lj} \right| > \varepsilon\right].$$

If $\varphi_1(1) < 1$ we can for $n$ sufficiently large directly apply Lemma (3.2) and get for some $\beta > \varphi_1(1)$

$$\begin{aligned} P\left[\left\| \sum_{j=1}^{r} X_{n,j} \right\| > \varepsilon\right] &< (\beta - \varphi_1(1))^{-1} \\ &\quad \cdot P\left[\left| (\sigma n)^{-1} \sum_{j=1}^{r} \sum_{l=1}^{n} \xi_{lj} \right| > \varepsilon/2\right] \\ &\leqq (\beta - \varphi_1(1))^{-1} 4\varepsilon^{-2} E[X_n^2(1, r/n)]. \end{aligned}$$

Since $\lim_{n\to\infty} \max_{1\leqq r\leqq [nh]} E[X_n^2(1, r/n)] = h$ we conclude

$$\overline{\lim_{n\to\infty}} \max_{1\leqq r\leqq [nh]} P\left[\left\| \sum_{j=1}^{r} X_{n,j} \right\| > \varepsilon\right] \leqq ch$$

for a constant $c$. From this assumption (4) follows in the case considered. If $\varphi_1(1) = 1$ we split the sum at first into subsums in the same way as we did in the inequality preceding (3.4) and apply (3.2) to the subsums. Finally since the assumptions of Theorem (3.4) can be verified following the same pattern our result holds in case $\varphi_2(1) = 1$ too. ⌟

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